

Fall 2025.

**Physics 501. Methods of Theoretical Physics.
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Homework # 2.

Due to September 18th, 2025.

Read the corresponding chapters. If you forgot anything necessary from your core math courses, find the information in the textbook, online or whatever works for you. Solve the problems. It should be done in a very detailed and logical way. You need to understand every step of your solution.

Infinite Series Method. Use the infinite series method to solve the differential equations. Find the indicial equation, series coefficients, and, if it is possible, a general form for the series.

- a) $y'' - y = 0$;
- b) $y'' - xy' - y = 0$;
- c) $y'' - x^2y' + y = 0$;
- d) $(1 + x^2)y'' - y' + y = 0$;

Frobenius Method. Use Frobenius method to find both linearly independent solutions of the following equations:

Two distinct roots of the indicial equation and $r_1 - r_2 \notin \mathbb{Z}$. $y_1(x) = \sum_{n=0}^{\infty} a_n(x - x_0)^{n+r_1}$, and $y_2(x) = \sum_{n=0}^{\infty} b_n(x - x_0)^{n+r_2}$. Write down the answer in the general form.

- e) $2xy'' + 3y' - y = 0$;
- f) $2x^2y'' + xy' - (3 - 2x^2)y = 0$;
- g) $2xy'' + (1 + x)y' + y = 0$;

Two distinct roots of the indicial equation, and $r_1 - r_2 \in \mathbb{Z}$. If r_1 is the larger root, $y_1(x) = \sum_{n=0}^{\infty} a_n(x - x_0)^{n+r_1}$. Try the smaller root of the indicial equation r_2 . Sometimes it works even though $r_1 - r_2 = N$, where N is a natural number. You might recognize the series as Taylor series of a composition of some elementary functions.

- h) $xy'' + 2y' - 4xy = 0$;
- i) $xy'' - y' + 4x^3y = 0$;

When the indicial equation has repeated roots, $r_1 = r_2 = r$, two linearly independent solutions have the following form: $y_1(x) = \sum_{n=0}^{\infty} a_n(x - x_0)^{n+r}$, and $y_2(x) = y_1(x) \ln(x - x_0) + \sum_{n=1}^{\infty} b_n(x - x_0)^{n+r}$.

j) $x^2 y'' - xy' + (1 + x^2)y = 0$;

k) $x^2 y'' + (x^2 - 3x)y' + 4y = 0$;

When $r_1 - r_2 = N$, where N is a natural number, there might be a logarithmic term in the second solution.

$y_1(x) = \sum_{n=0}^{\infty} a_n(x - x_0)^{n+r_1}$ (r_1 is the greater root!), and $y_2(x) = Cy_1(x) \ln(x - x_0) + \sum_{n=0}^{\infty} b_n(x - x_0)^{n+r_2}$

l) $x^2 y'' + (2x^2 - 3x)y' + 3y = 0$.